

Universal Surgical Alignment: Leveraging Canonical Textual References for Zero-Shot Workflow Recognition.

Archit Goswami

Can we develop structured canonical text descriptions for surgical workflows to establish a universal, machine-readable procedural reference?



Can we then leverage hierarchical video-language alignment to automatically synchronize surgical video segments with these canonical steps to enable zero-shot procedural recognition?

Why?

- This helps us build checkpoints
 - Checkpoints indicate progress
 - Steps missed indicate issues or/and special circumstances
 - Most importantly: **Helps us establish surgical goals and if they are being hit.**

Phases of Surgery

4 phases of surgery:

1. access
2. remove/repair/reconstruct
3. secure
4. repair (?)

How?

(how do we categorize meaningfully)

Current research ¹ proposes - Multi-level Semantic Scene Understanding (MSSU) proposes a standard hierarchy:

1. Phase
2. Step
3. Action
4. Instrument
5. Anatomy

Breaking down all text and fitting all elements into these categories. Similar to Intents, Actions, and Entities in conversational flow NLP.

How?

(how do we categorize meaningfully)

We could also utilize Action-Driven Digital Twins (ActDTs)^[2], which creates subject-action-object triplets in JSON. We could map those to 2 of the 4 categories to better understand what is going on when.

[2]: <https://arxiv.org/abs/2606.17298>

How?

(analyzing videos and associating canonical steps to each section)

OR3 from “Reasoning Text-to-Video Retrieval for Operating Room Clips via Action-Driven Digital Twins”^[2] maybe helpful to understand but does the inverse (helps retrieve certain sections of a video based on an idea/occurrence of an event). Might be useful in tagging but unsure.

[2]: <https://arxiv.org/abs/2606.17298>

Issues: Zero-Shot Recognition

Idea to catalogue never before seen video to known canonical text in order to improve the recognition of steps and recognitions of surgery (I.E. width and depth) over time.

HecVL model works on:

- at clip-level, atomic actions using transcribed audio texts;
- at phase-level, conceptual text summaries;
- at video-level, overall abstract text of the surgical procedure.

Issues: Skipping steps

How do we deal with steps that we need to skip?

Also how do we ensure we “stay on track” when the rest of the video is still the valid procedure but just that one step is missing.

SurgPLAN++⁴ deals with staying on track while skipping steps.

[4]: <https://arxiv.org/abs/2409.12467>

Notes: StepAL

Need to read through Nisarg's paper to understand how step wise analysis done by him can assist with the second part of the research question.

Step aware active learning for surgical videos captures distribution of steps across videos, deals with providing data for the second part of the research question.

Using Step-aware Feature Representation (SFR) to capture the distribution of steps across the entire video so the model doesn't lose context

Lego Recipe Building

Can we think of steps as interchangeable between surgeries

So steps within phases might be worth logging and using interchangeably?

How?

(how do we categorize meaningfully)

First we would have to collect all text data on procedure and then categorise it. But that raises the question: how do we catalogue meaningfully?